

Exercise Sheet

Task 1 – Hamming Codes

- a) Use the table to encode these bitstrings: $(10101110001)_2$, $(11011010011)_2$
- b) Use the table to decide at which position an error occurred: $(101101101010101)_2$, $(1110101110101011)_2$

Bit: position	8	4	2	1	Bit: value
3	X	X			
5	X		X		
6	X			X	
7	X				
9		X	X		
10		X		X	
11		X			
12			X	X	
13			X		
14				X	
15					
Parity bits (computed):					
Parity bits (recieved):					

Task 2 – Bit-PLRU

Have a look at the following sequence of accesses:

0xAB47, 0xB759, 0xC7D5, 0xB759, 0xD8AA, 0xEBFF, 0xC7D5, 0xFFFF1,
0xC7D5, 0xD8AA, 0xEBFF, 0xAB47, 0x7A39, 0xC7D5

- a) Fill in the Cache table for a fully associative cache that has 4 cache lines and uses Bit-PLRU as replacement policy.
Decide how many misses occur and of what type these misses are.
- b) What is the minimum number of misses for this access sequence in a cache with 4 lines? (Hint: Optimal Replacement Policy)

Access	Line 1	Status	Line 2	Status	Line 3	Status	Line 4	Status	Hit/Miss(-type)
0xAB47									
0xB759									
0xC7D5									
0xB759									
0xD8AA									
0xEBFF									
0xC7D5									
0xFFF1									
0xC7D5									
0xD8AA									
0xEBFF									
0xAB47									
0x7A39									
0xC7D5									